Molecular Biology & Genetics
2026

Dr. Rula Abdul Ghani

Molecular Biology: Course Outline

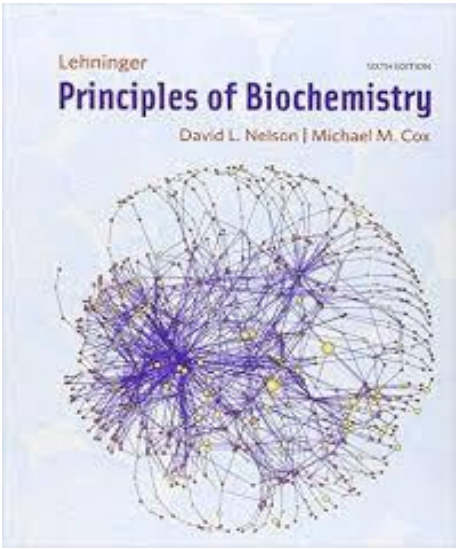
Genes and chromosomes

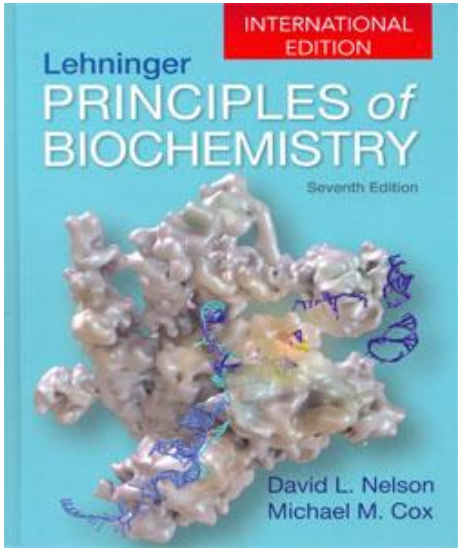
DNA metabolism

RNA metabolism

Protein metabolism

Techniques



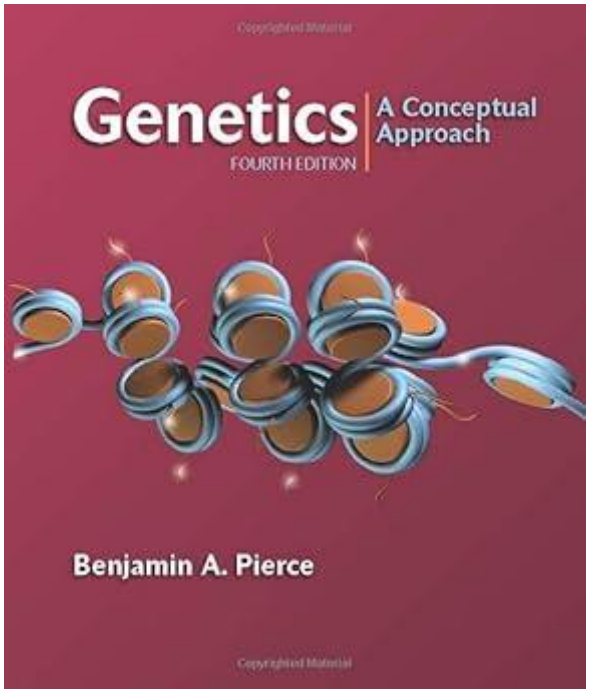


Text Book :

Lehninger Principles of Biochemistry 6th / 7th edition

Genetics
Course Outline

- Introduction
- Mendelian and Non Mendelian genetics.
- Modes of Inheritance
- Single gene disorders
- Chromosomal disorders/abnormalities
- Cancer (oncogenes and tumor suppressors)
- Gene therapy



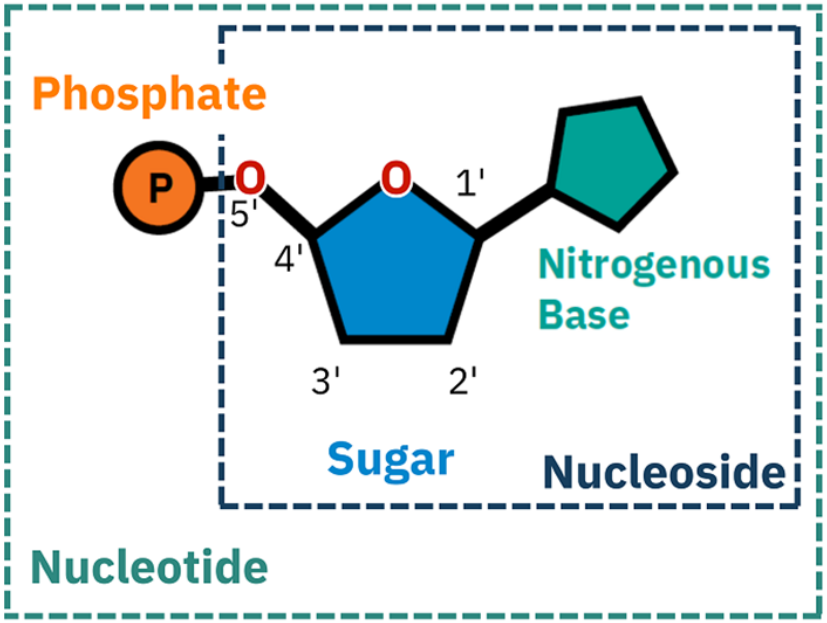
Text Book:

Genetics: A conceptual approach.

DNA

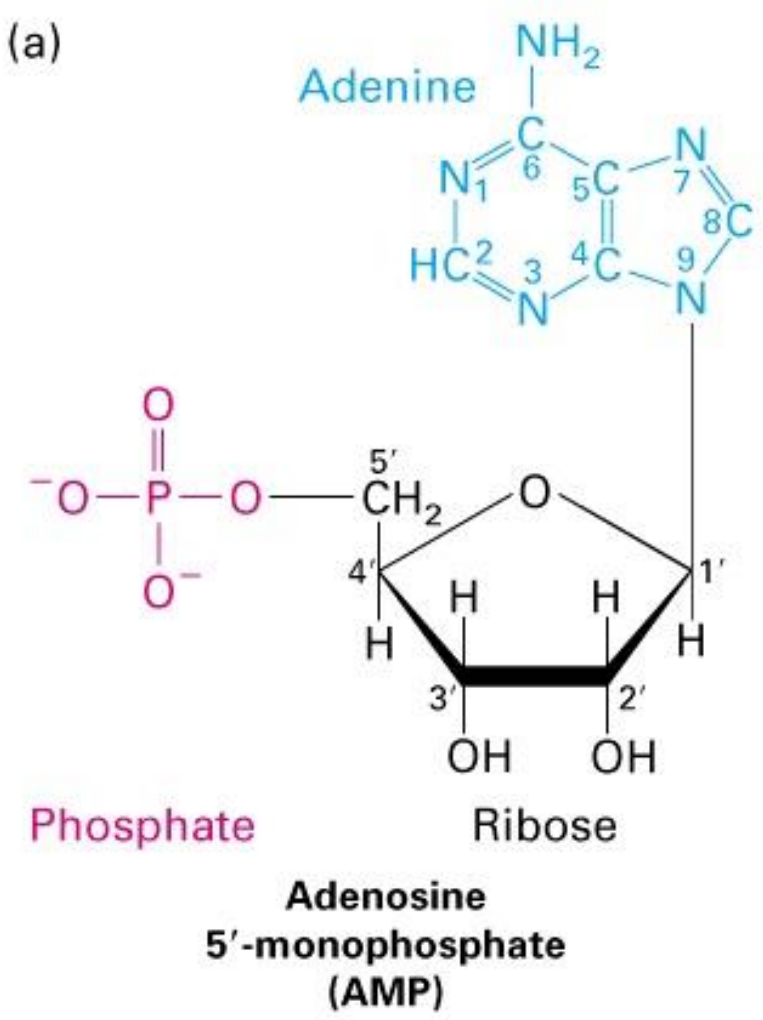
DNA, or deoxyribonucleic acid, is a molecule that carries the genetic instructions for the development, functioning, and reproduction of nearly all living organisms.

The building block of DNA is the nucleotide:

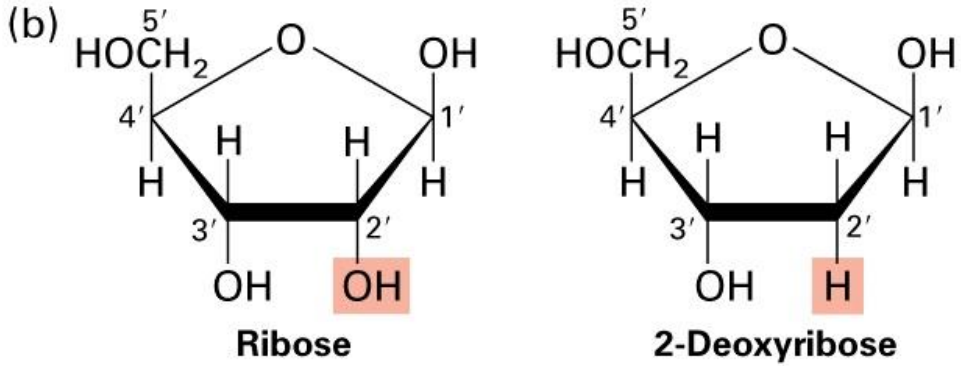


All nucleotides have a common structure

A nucleotide present in RNA



The pentoses in nucleic acids

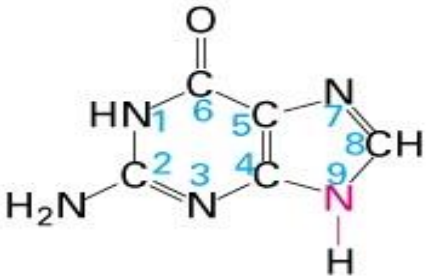


There are five Nitrogenous bases in nucleic acids

PURINES

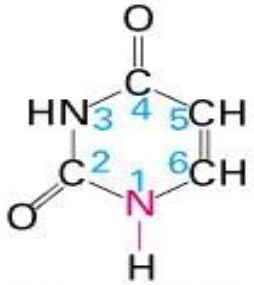


Adenine (A)



Guanine (G)

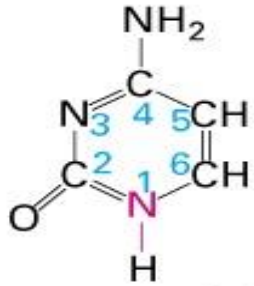
PYRIMIDINES



Uracil (U)



Thymine (T)



Cytosine (C)

A, G, T, C are present in DNA

A, G, U, C are present in RNA

Nitrogen 9 of purines and nitrogen 1 of pyrimidines are bonded to C1 of ribose or deoxy ribose

DNA Structure

The primary structure of DNA is the sequence of nucleotide bound together by the *phosphodiester bond*.

The secondary structure is the double-stranded helix.

The tertiary structure is the packing/ folding of DNA.

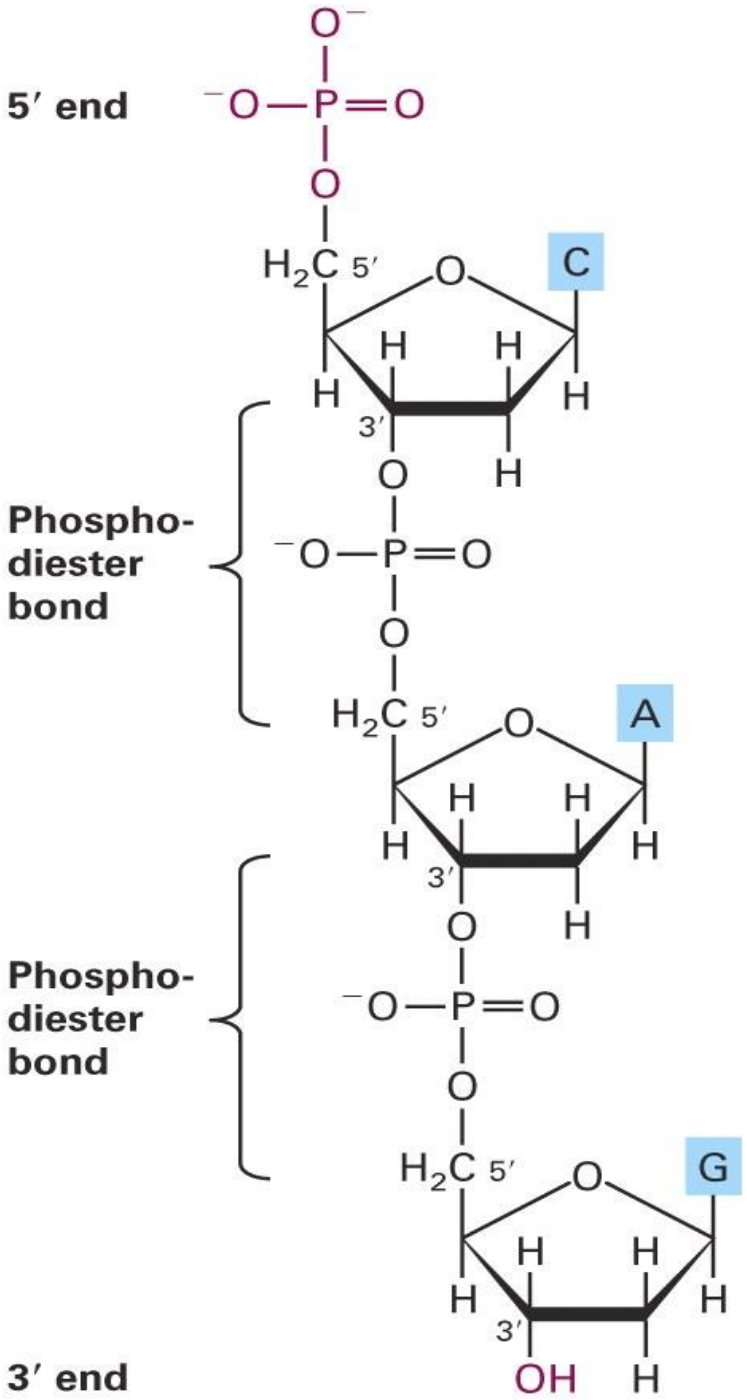
Nucleotide subunits are linked together by phosphodiester bonds

This nucleic acid chain is represented as (5’- CAG -3’)

always in 5’ to 3’ direction left to right.

5’ phosphate group on the 5’ of the sugar and a 3’ hydroxyl of the sugar

CAG

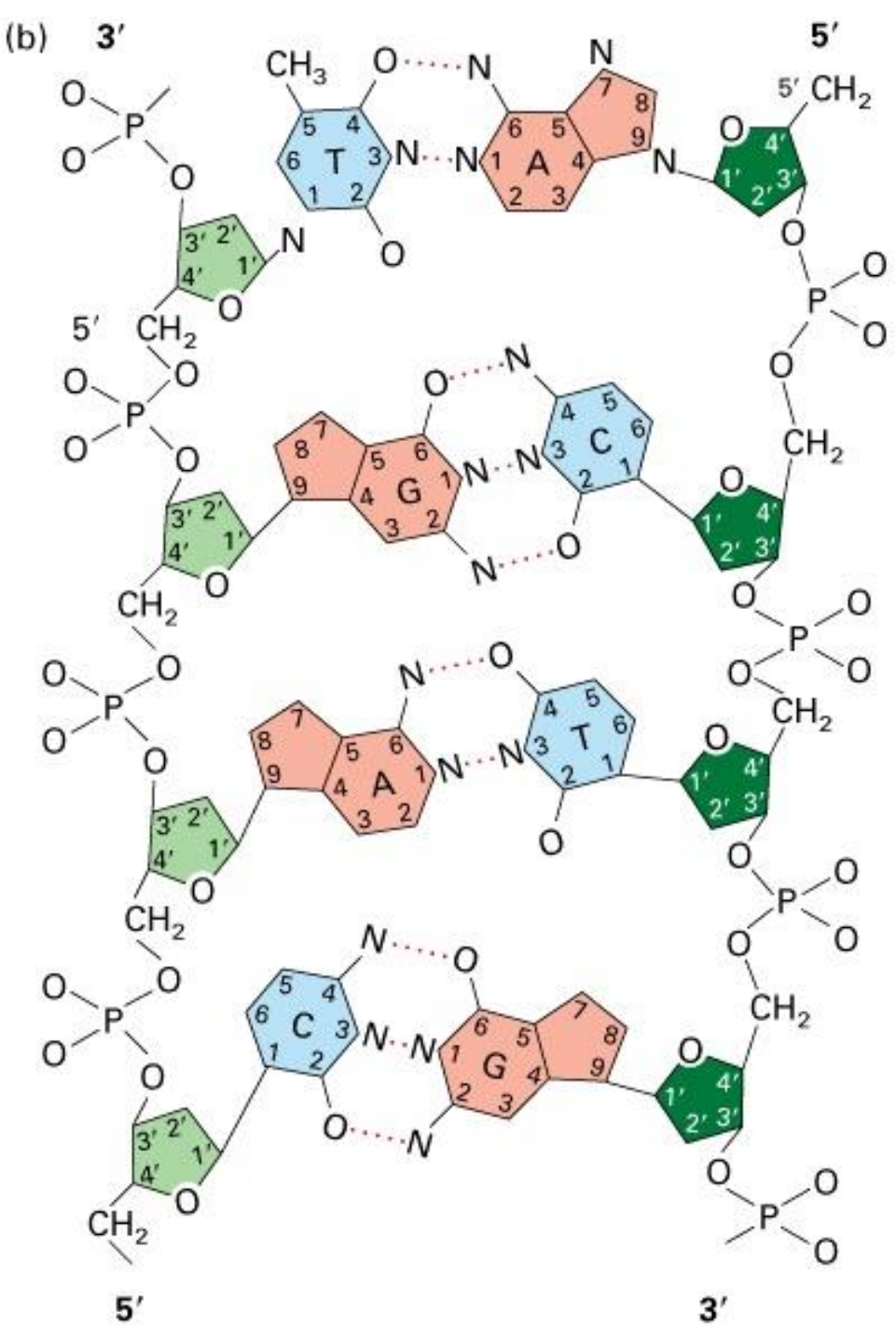


Native DNA is a double helix of complementary antiparallel chains held together by:

1- Hydrogen bonding between complementary base pairs (A-T or G-C)

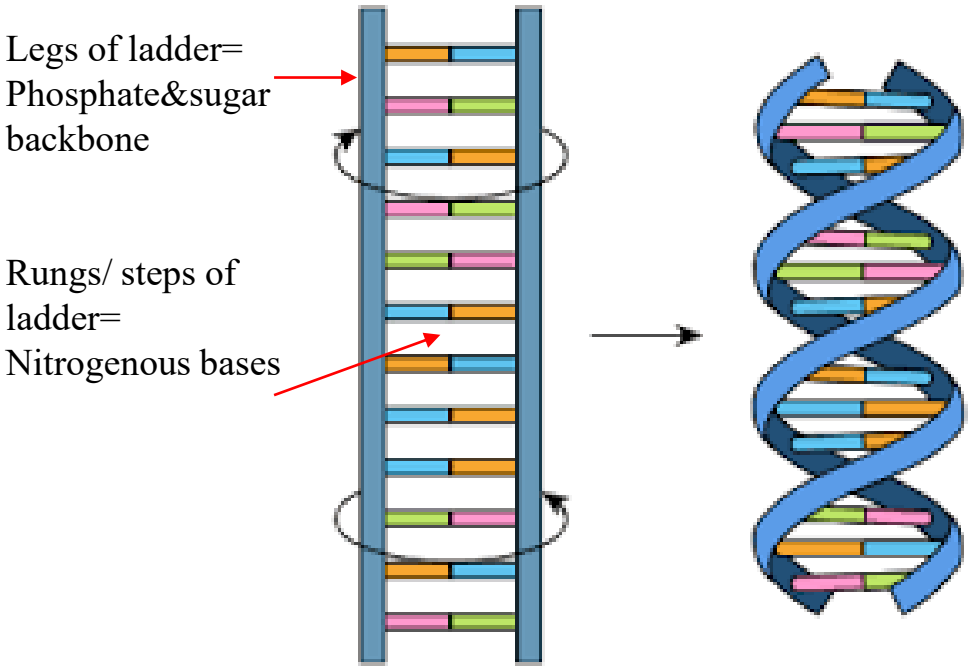
2- Hydrophobic interactions between planar bases/ stacked adjacent bases.

This contributes to the stability of the double helix.



DNA double helix

The Watson and Crick model describes DNA as a double helix, a twisted ladder structure.
The nitrogenous bases (A, T, C, G) pair in the center—Adenine with Thymine (A-T) and Guanine with Cytosine (G-C)—held together by hydrogen bonds.





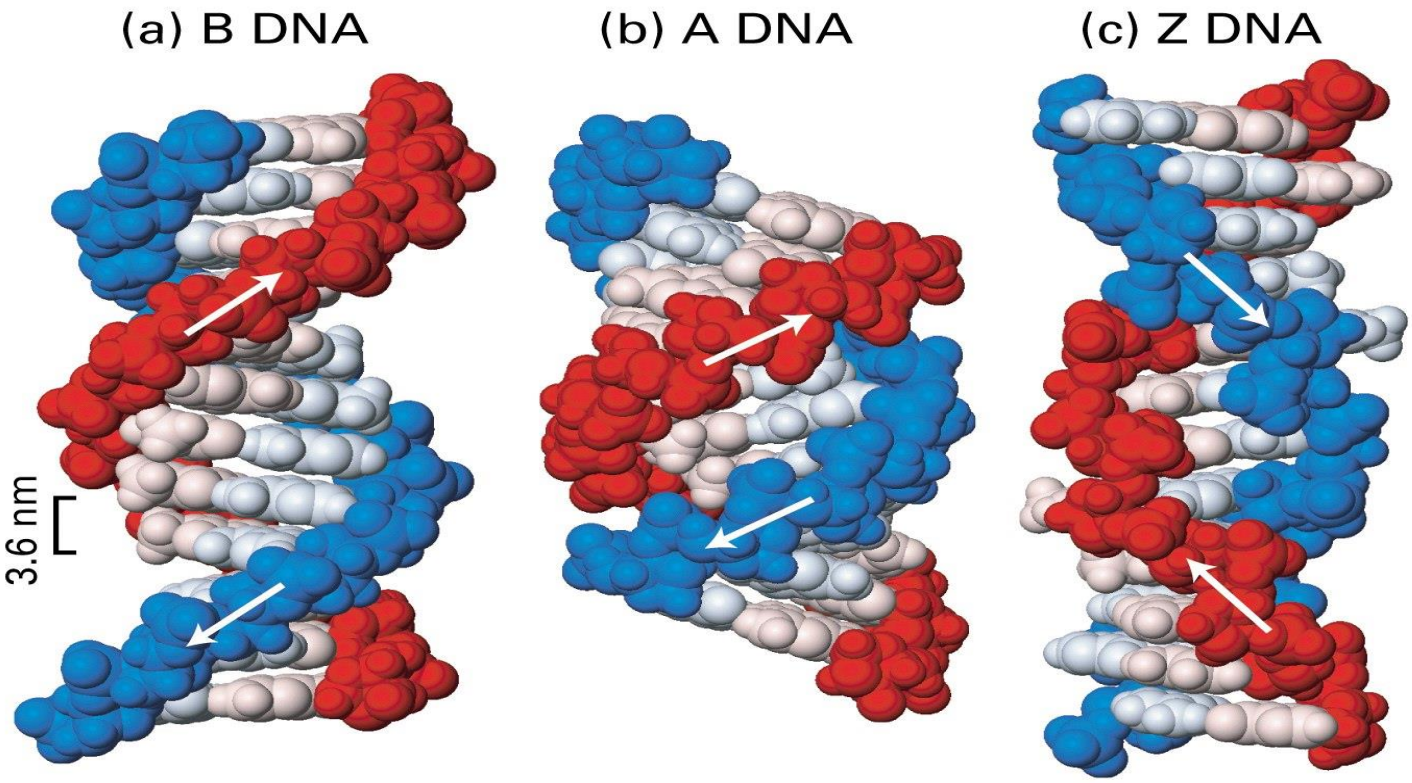
DNA helix

DNA coiled in the form of double helix , with both strands coiling around an axis.

The most stable form of DNA is B form, right handed helix.

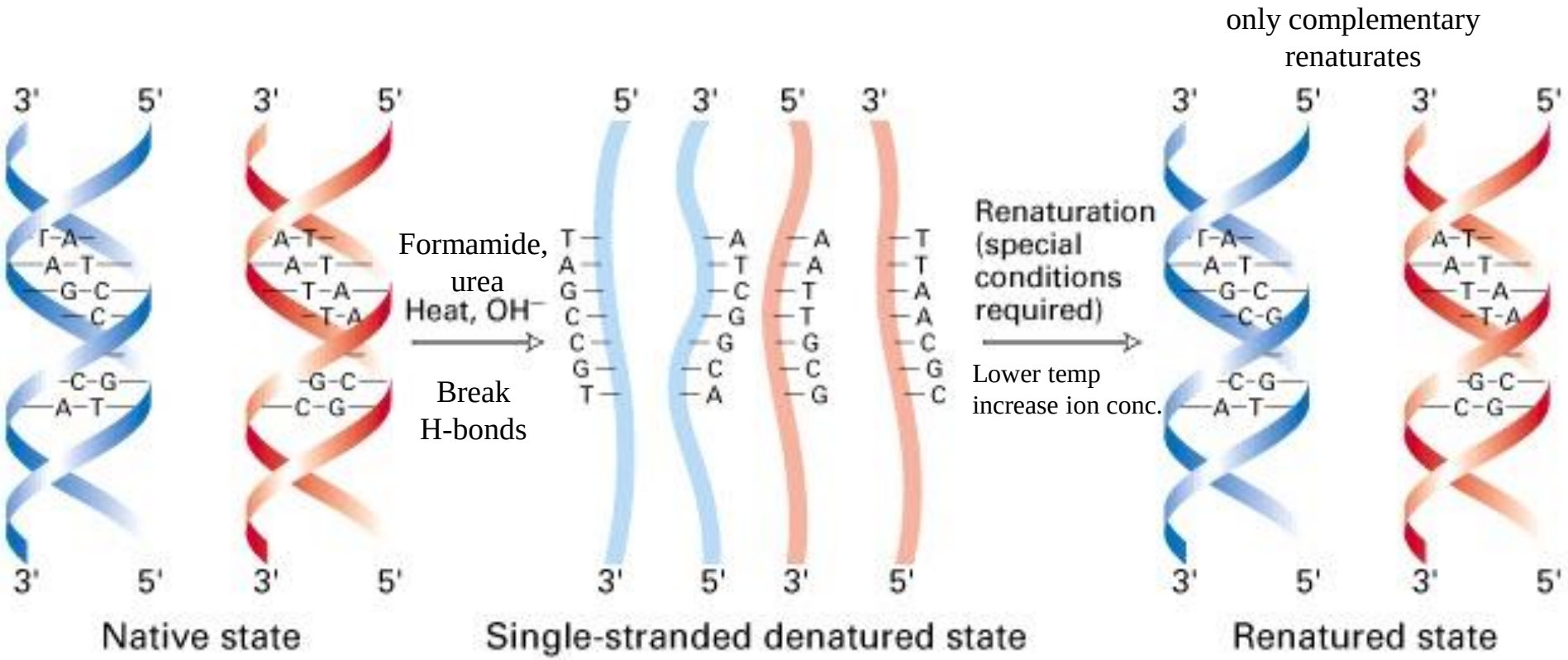
	B DNA	A DNA	Z DNA
Helix sense	Right-handed	right	left
Mean bp/turn	10.5	10	12

Further, axis coiling upon itself produces DNA supercoiling



DNA can undergo reversible strand separation

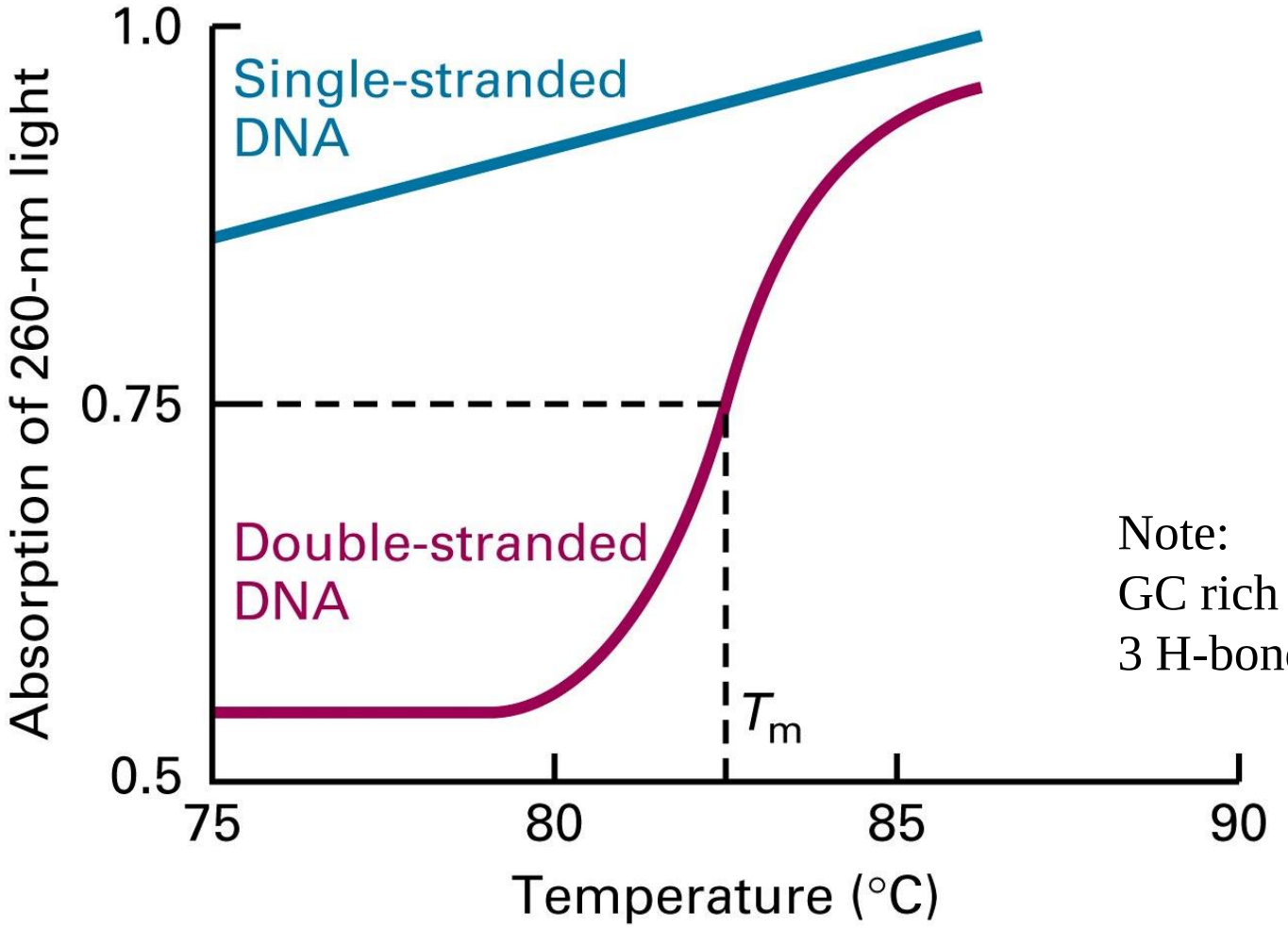
denaturation/ melting = Unwinding and separation of the double strand



DNA denaturation changes its absorption to UV at 260 nm (used to measure DNA concentration.

DNA denaturation extent dependent on time, DNA conc., ionic content of soln.

Analysis of DNA denaturation



Note:
GC rich higher Tm:
3 H-bonds to break.

T_m= temperature at which ½ the bases in a dsDNA sample have denatured

- DNA denaturation and renaturation is the

basis of nucleic acid hybridization and PCR = a powerful technique in Molecular studies.

Prokaryotic vs Eukaryotic DNA

	Prokaryotes	Eukaryotes
Location	Freely in nucleoid region in the cytoplasm	Membrane-bound nucleus
Shape	Typically a single, circular	Multiple Linear chromosome
Packeging	Naked" DNA, not associated with histones	Tightly coiled around histone proteins to form chromatin
Plasmids	Often contains extra-chromosomal plasmids Generally	lacks plasmids (but organelles like mitochondria can have their own chromosomes)
Genome size	Generally much smaller and more compact.	Significantly larger and more complex

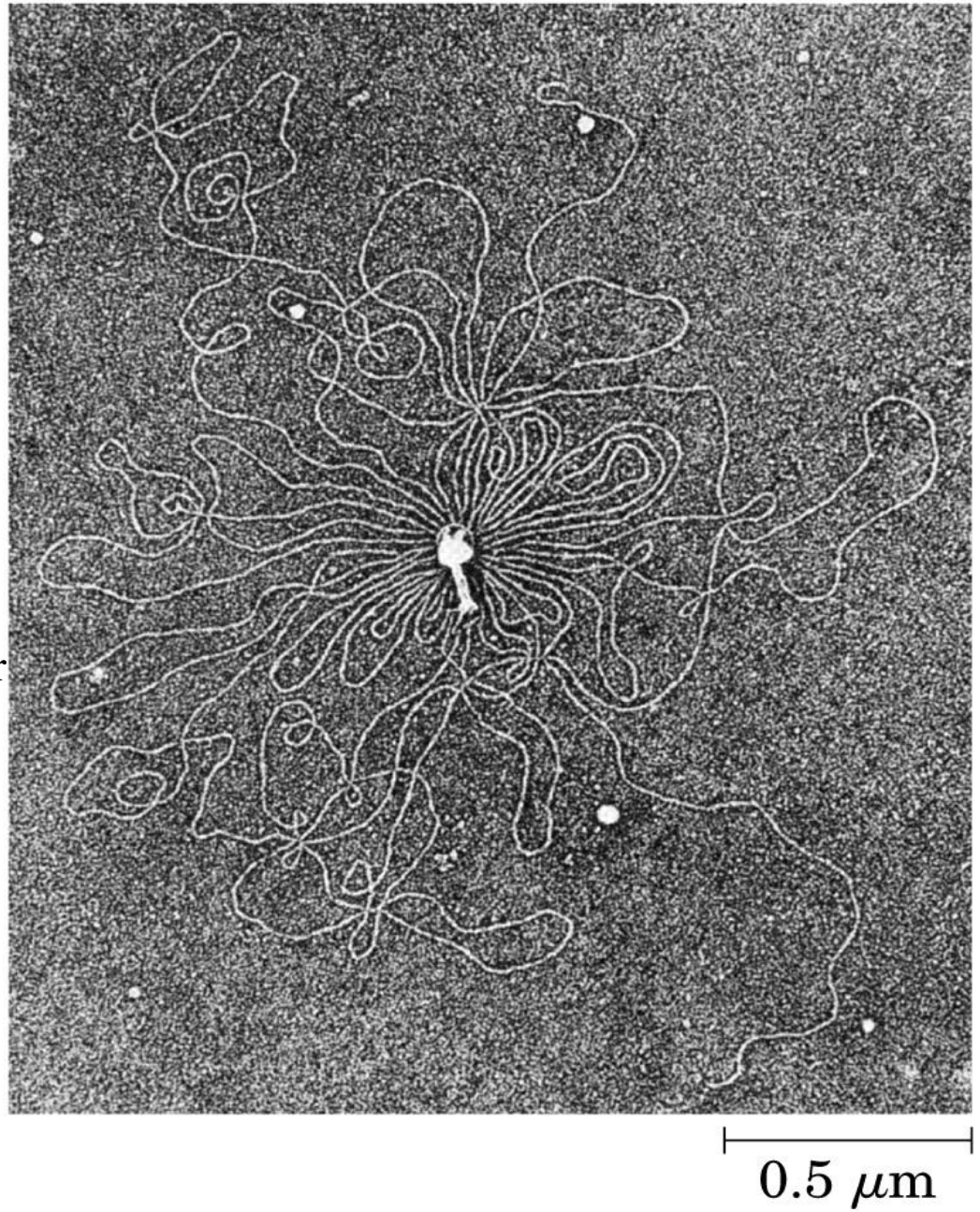
DNA in Virus

DNA much longer than the virus or viral packege that contain it.

Bacteriophage T2 lysed in distilled water allowing DNA to spread on water.

- protein coat surrounded by its single linear DNA .

- Normally in undamaged T2, DNA is packaged in the phage head.

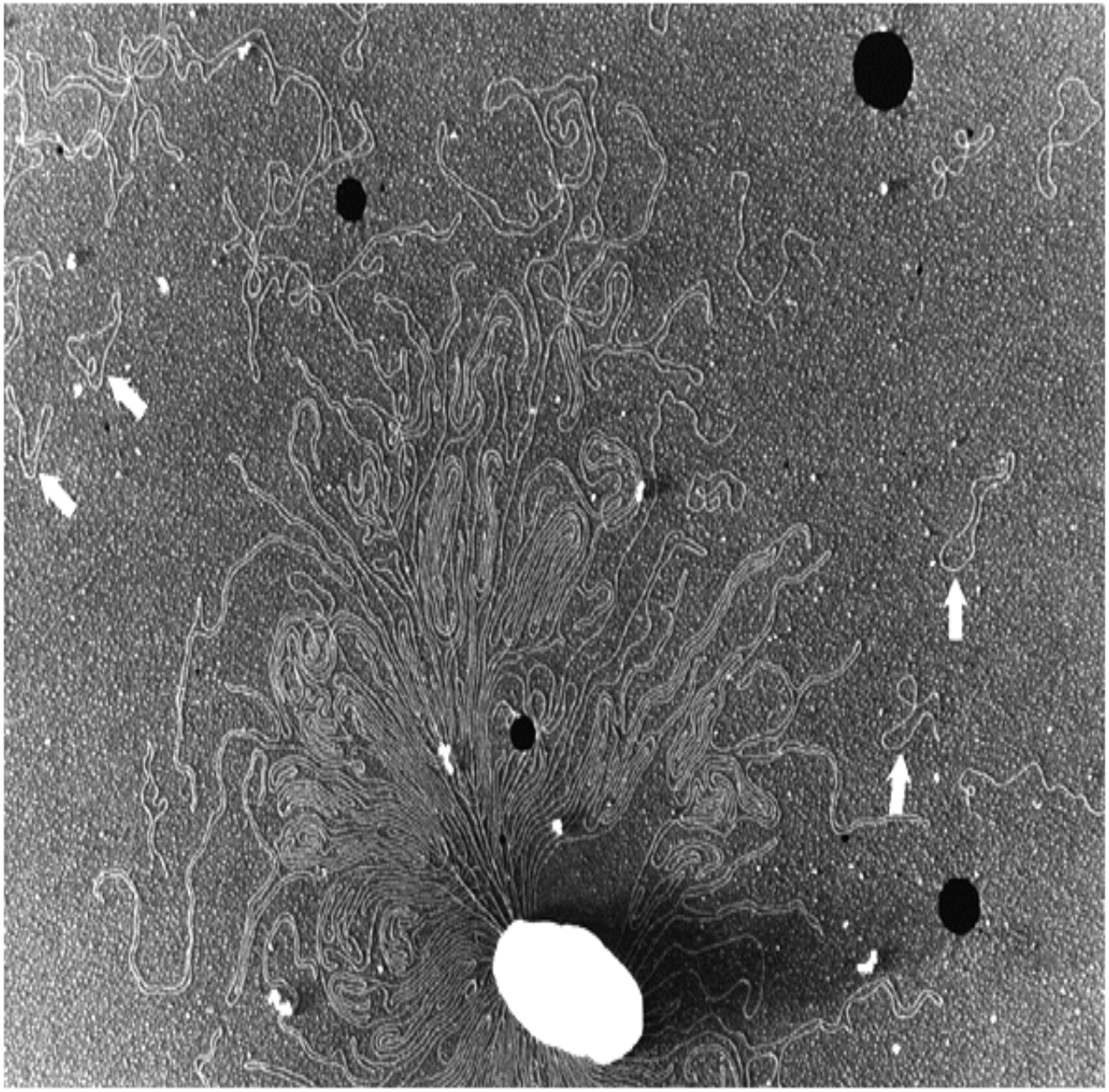


DNA in Bacteria:

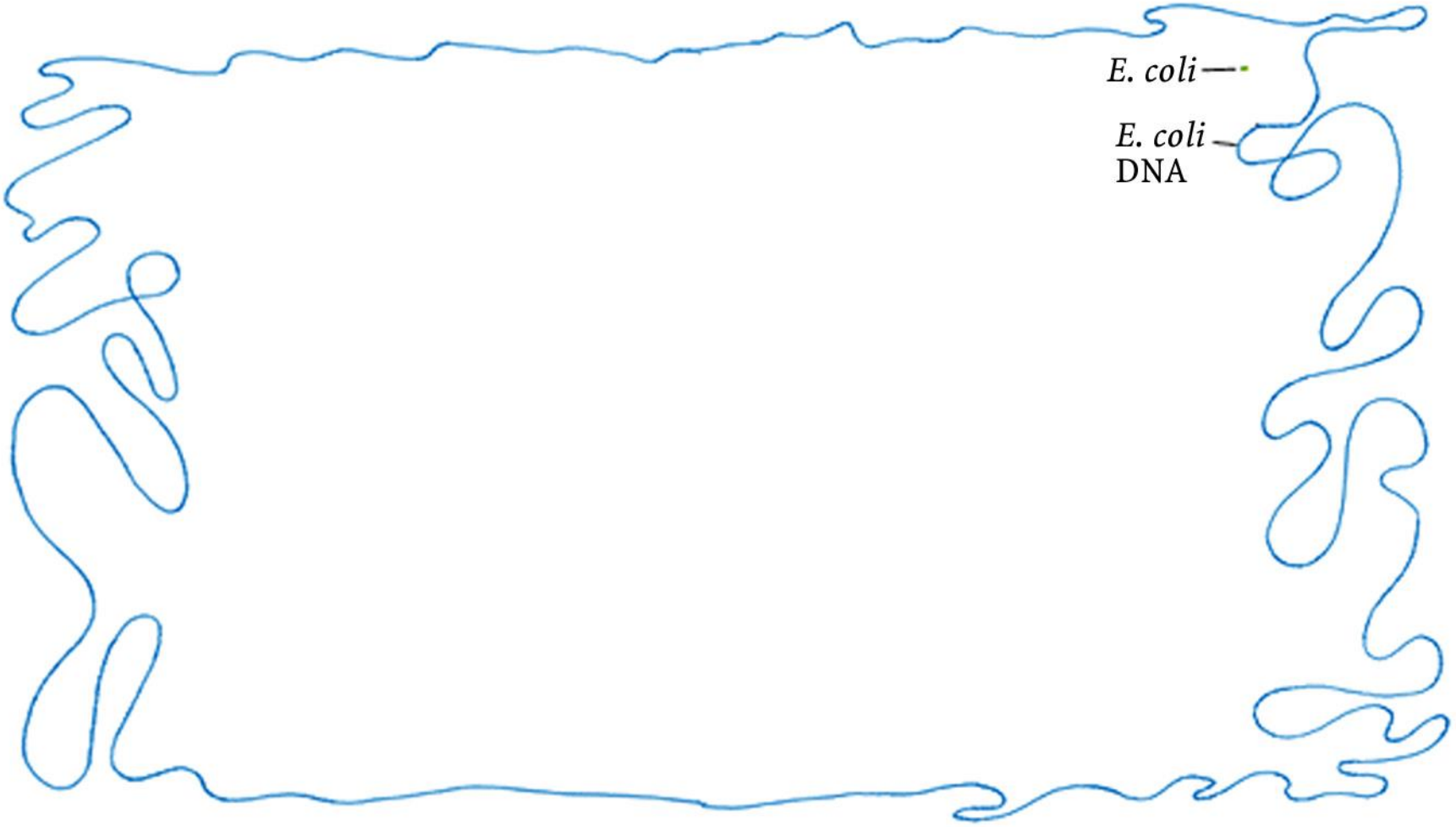
- E. coli 100> DNA virus.

-One large circular DNA in <u>nucleoid + plasmids</u>

DNA from a lysed E coli→



The length of the E coli chromosome(1.7mm) relative to the length of cell (2um).

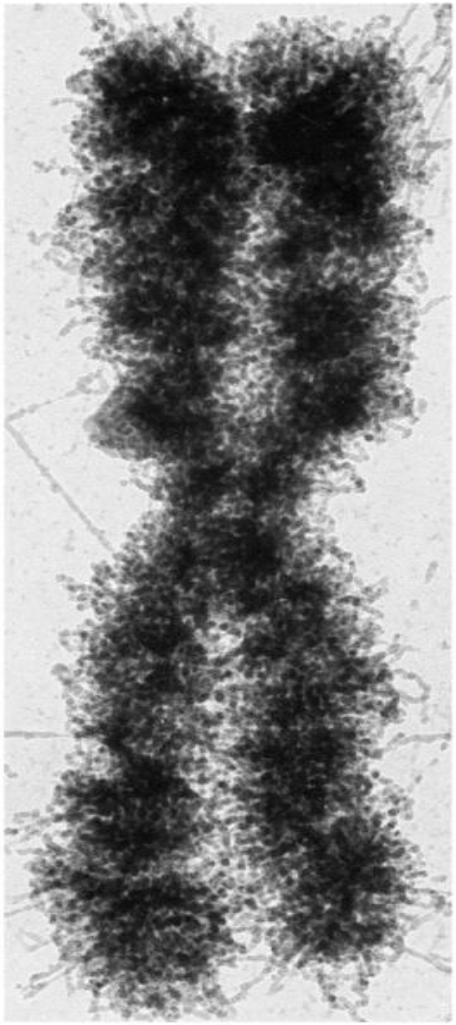


Eukaryotic DNA:

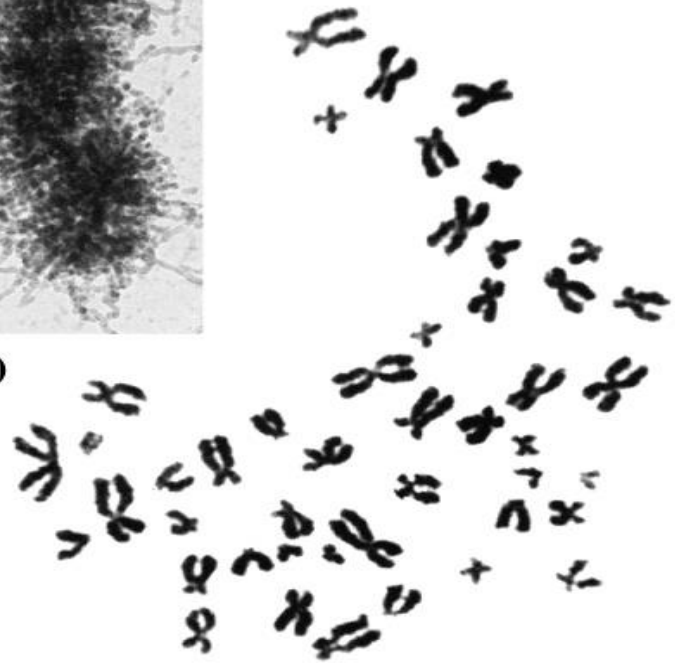
The DNA of human genome:

46 chromosomes in every somatic cell.
If placed end to end = 2m DNA.
2m in 0.006nm diameter nucleus!!!

Adult human body= 10_{13} cell
Length of DNA = 2 x 10_{13} m
the DNA can stretch to the sun and back 50 times.



(a)



(b)

mitochondrial DNA (mtDNA): 2-10 copies mtDNA /each mitochondria

The size, coding capacity vary greatly among organisms, human mtDNA among the smallest containing 16,569bp <<< yeast, plant

Encode proteins of mitochondrial e-transport chain and ATP synthesis.

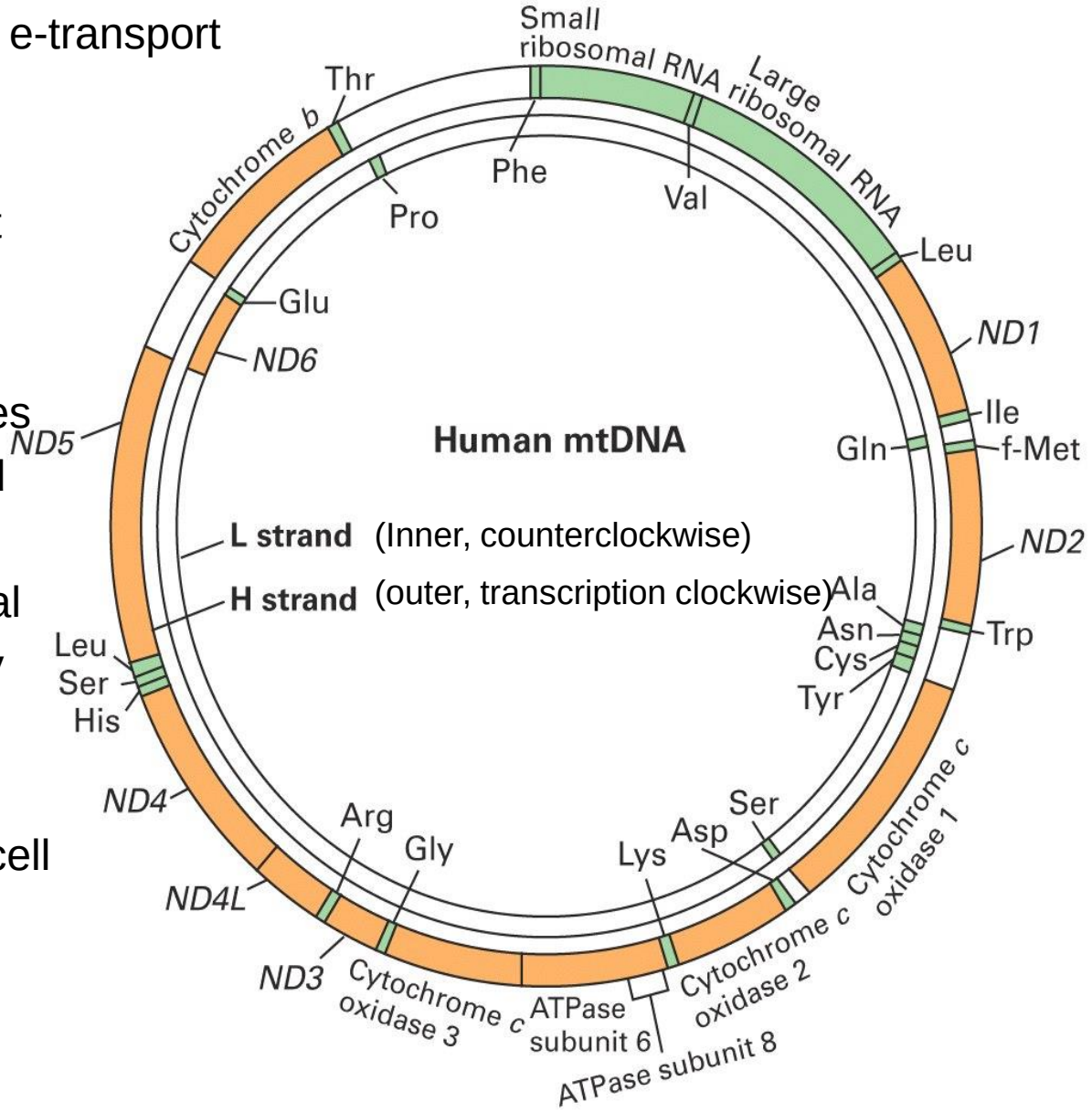
1) mammalian mtDNA in contrast to nuclear lacks introns and non coding regions.

2) products of mitochondrial genes are not exported, but synthesized on mitochondrial ribosomes.

--mt ribosomes resemble bacterial ribosomes in structure, sensitivity or resistance to drugs.

3) mtDNA divides each time the cell divides before cell division.

4) Mitochondrial genetic codes differs from nuclear.



Gene =

A portion of chromosome that determines a single specific character / phenotype/ visible property.

A segment of genetic material that determines /codes for a protein / enzyme:

One gene one enzyme-------One gene one polypeptide.

Modern biochemical definition:

All the DNA that encodes the primary sequence of some final gene product (polypeptide / RNA).

How many genes in a single chromosome?

E coli (prokaryotic) = 4,6 million bp

encodes 4300 gene

Human = 3.2 billion bp

encodes around 21000 genes

Introns:

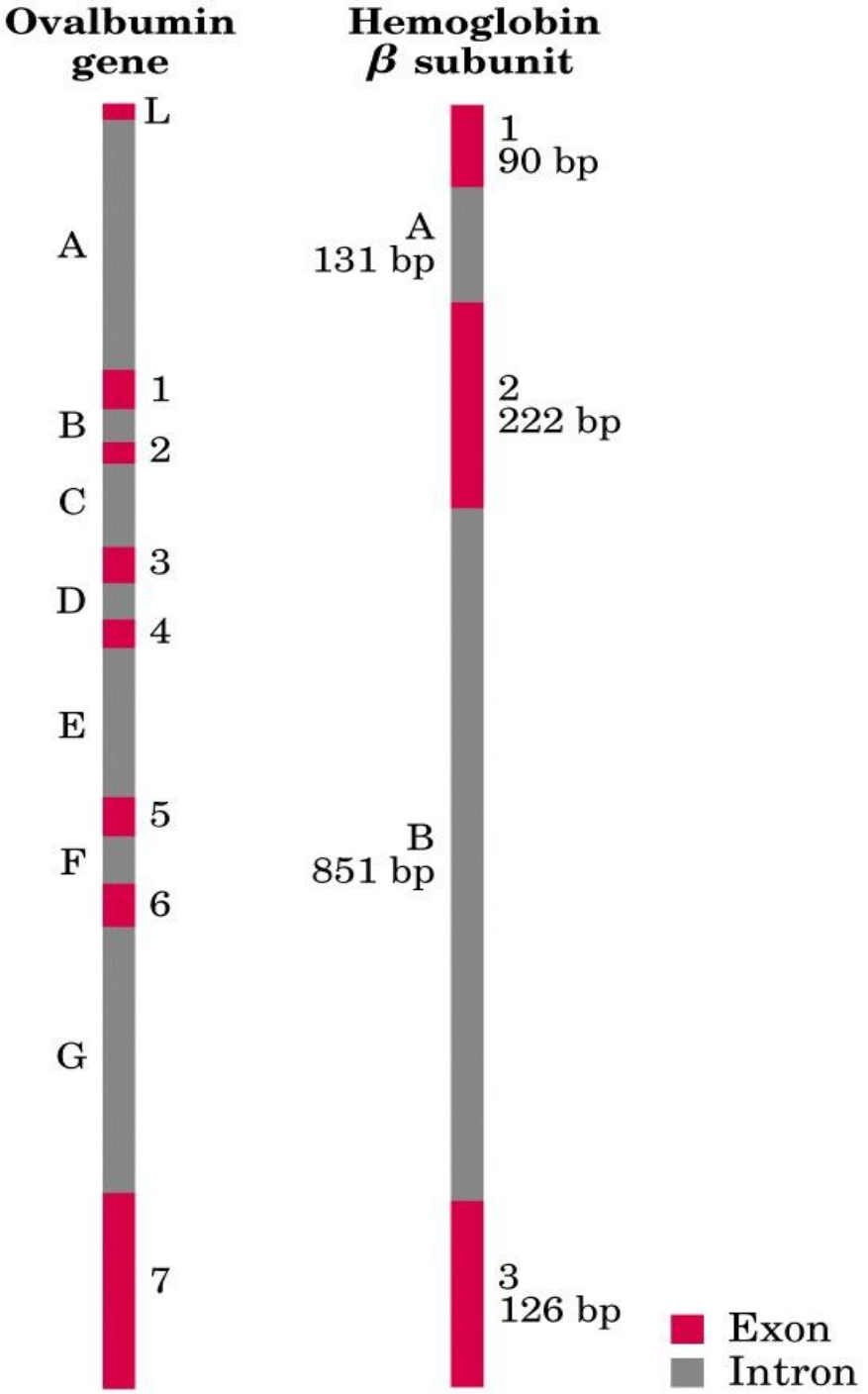
intervening sequences = nontranslated DNA segments in genes.

Exons:

a region of DNA within a gene transcribed to final (mRNA) molecule, rather than spliced out from transcribed RNA.

Introns in 2 eukaryotic genes:

Ovalbumin introns (7) much longer than exons (7+L) , introns=85% of the gene.

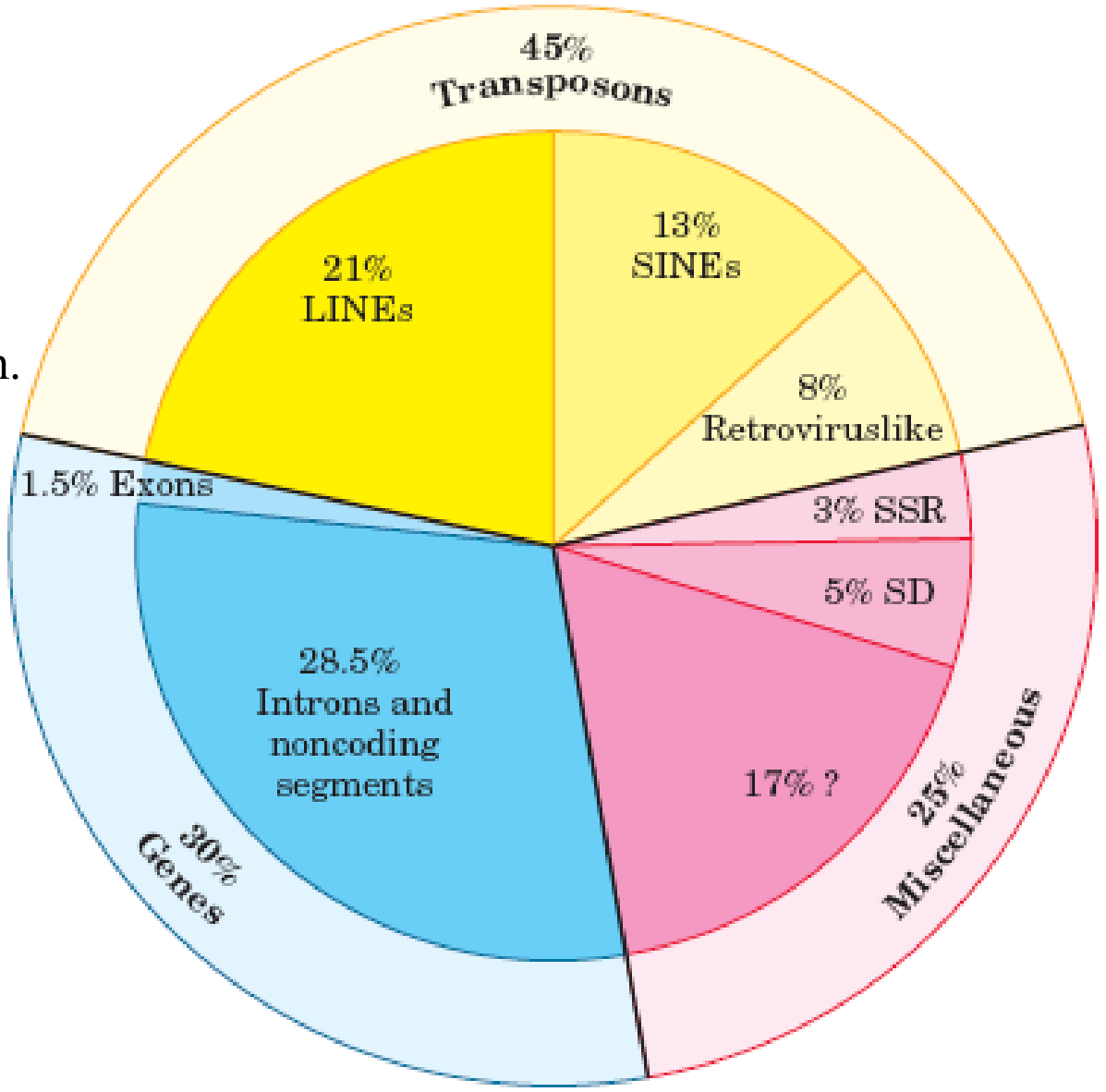


Types of sequences in human genome:

SINEs: Short interspersed elements 100-300bp long

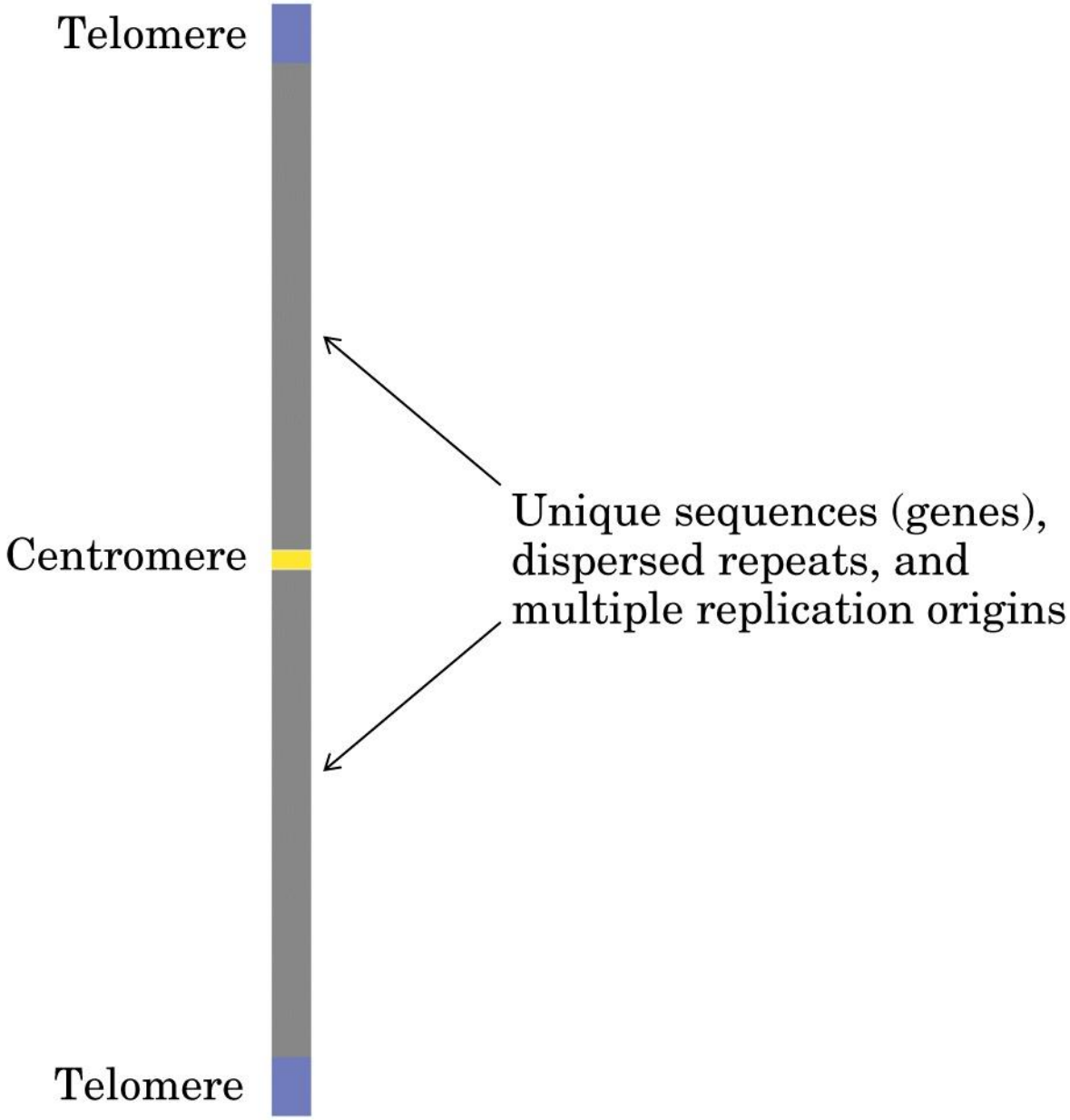
LINEs: Long interspersed elements 6-8kbp, encode a few genes that catalyze transposition.

SSR: simple sequence repeats < 10bp repeated millions times /cell



Centromeres:
a sequence functions during cell division as an attachment point during mitosis,
A=T rich sequence.

Telomeres:
sequence at the end of chromosomes that help stabilize chromosome

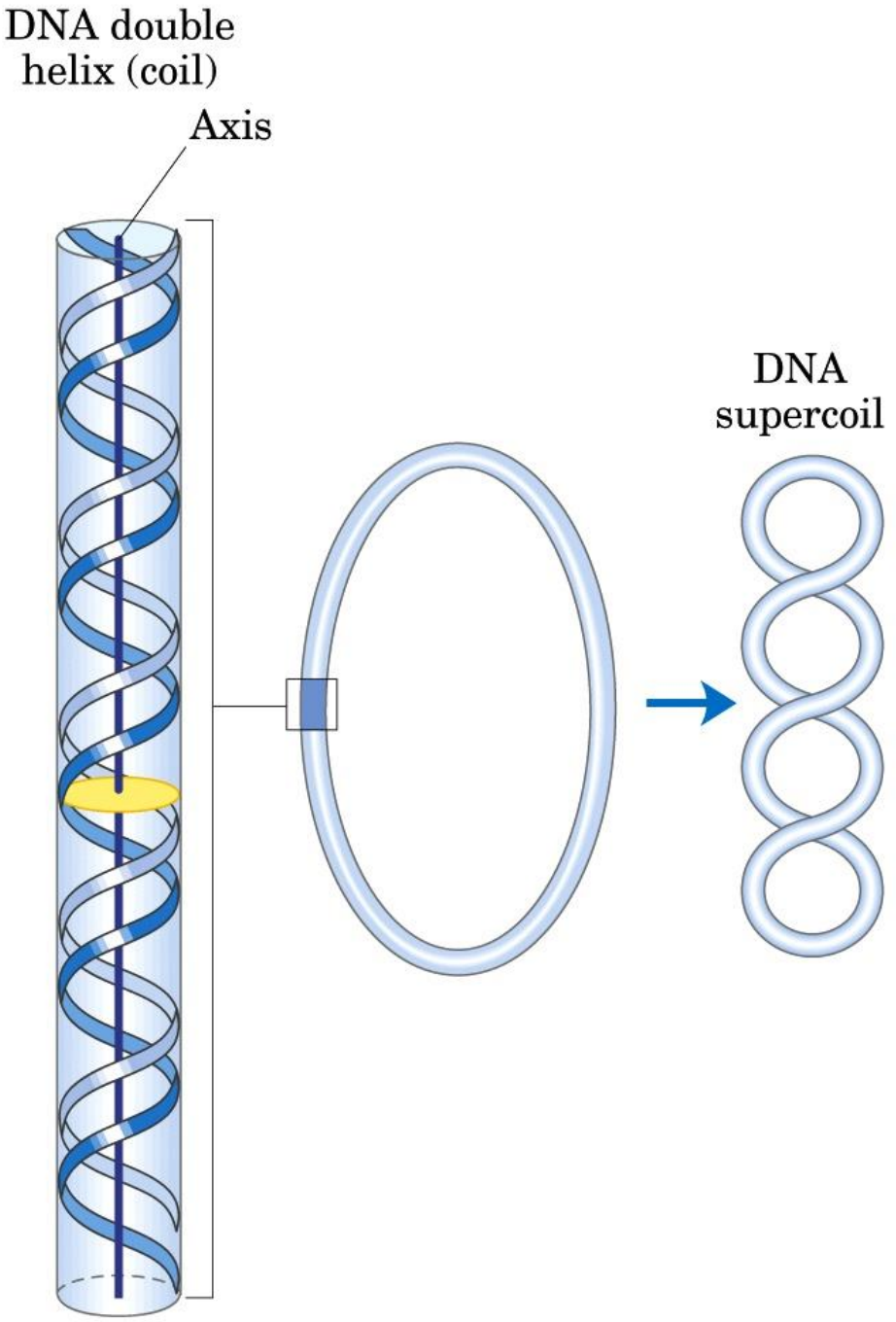


Supercoiling of DNA:

When the axis of DNA double helix is coiled on itself it forms a new helix (superhelix).

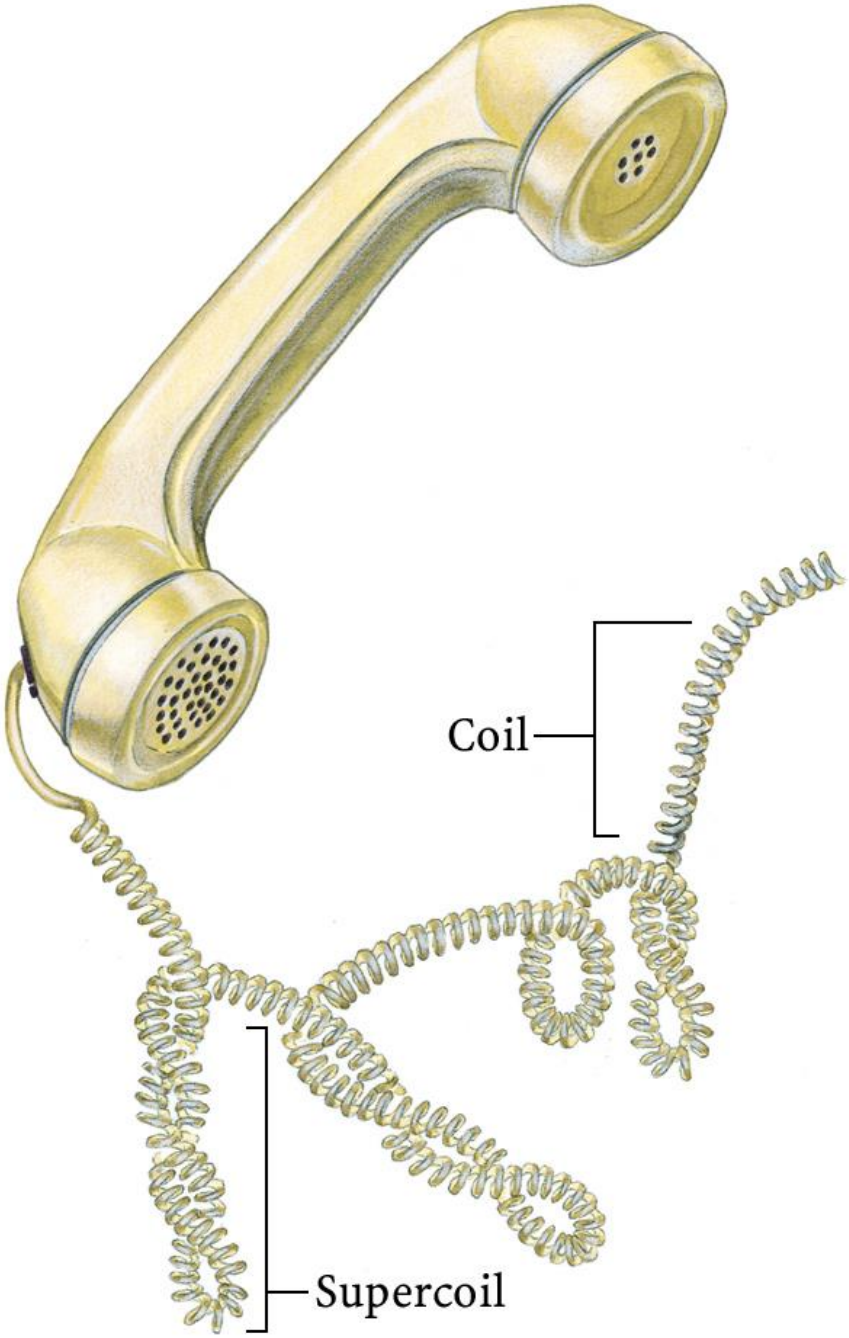
Superhelix = supercoil

The DNA in the **relaxed state**:
No net bending of the DNA axis upon itself.



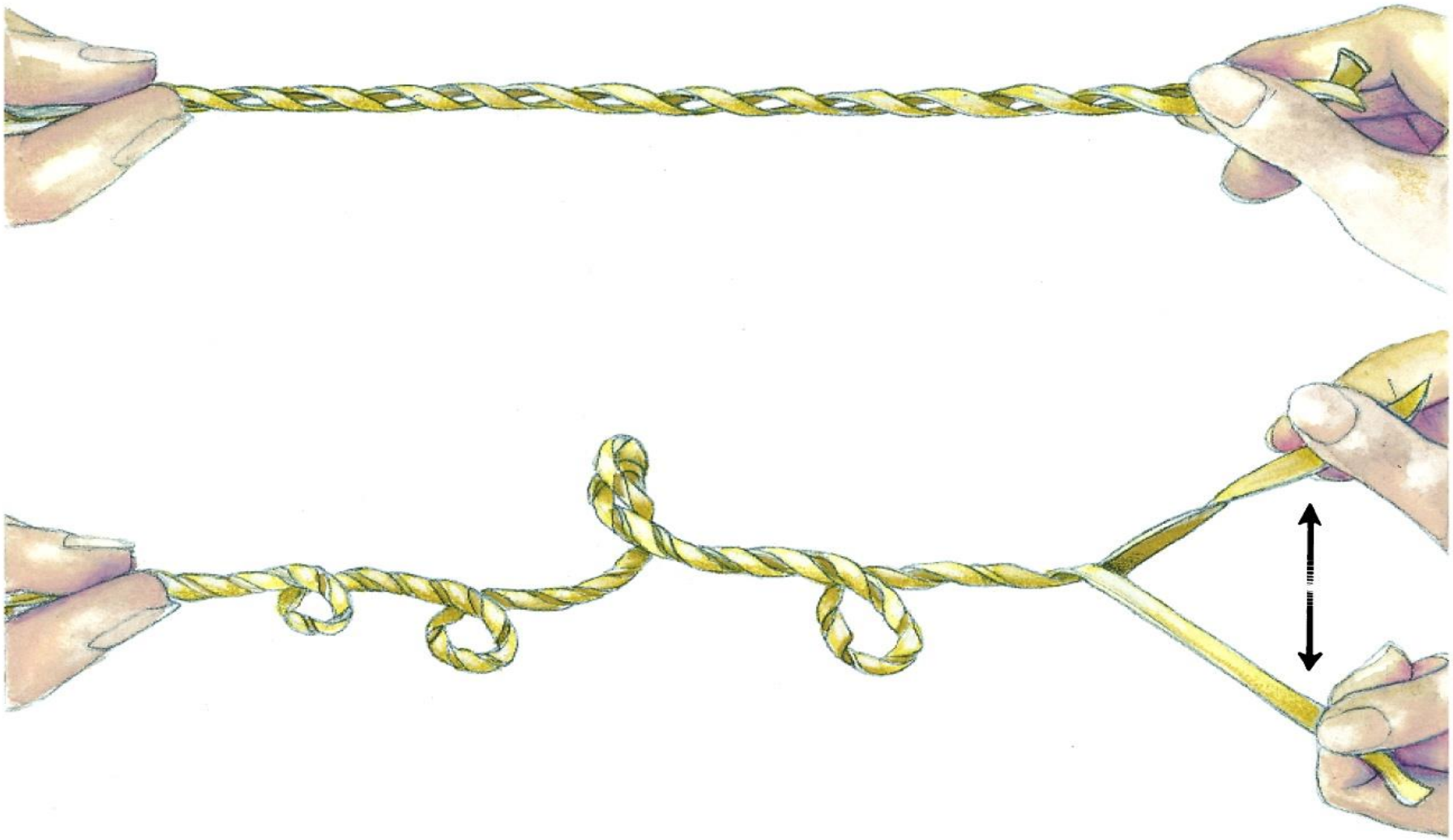
Supercoiling:

coiling of a coil



Supercoiling induced by separating the strands of a helical structure.

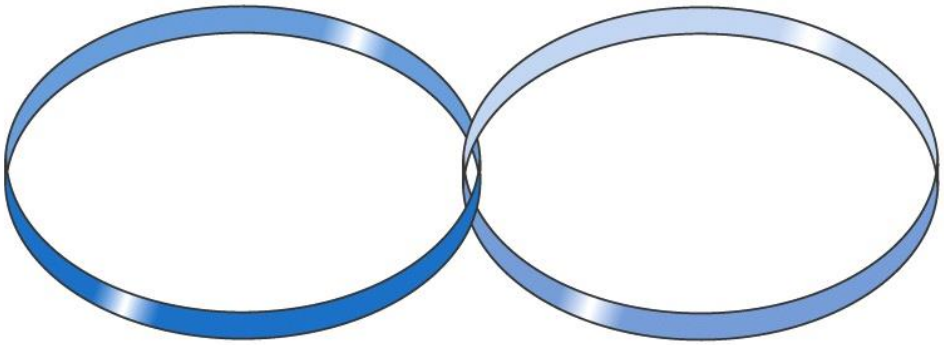
Unwinding and subsequent supercoiling occurs during replication, transcription, and protein binding.



Linking number Lk:

number of times the 2nd strand pierces / penetrates the surface

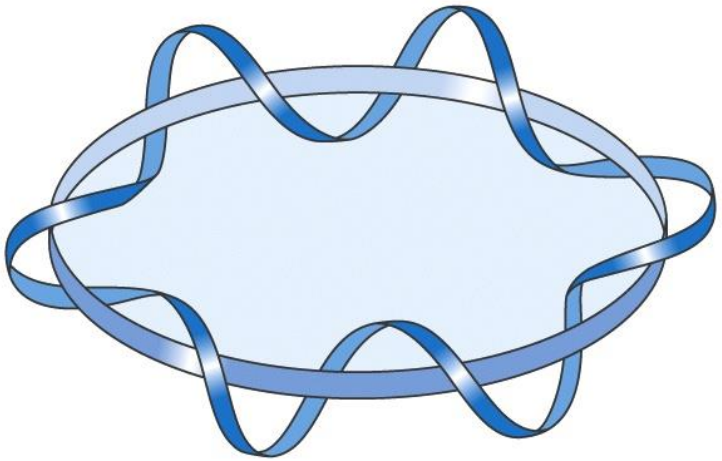
To change LK you must break one strand.



$Lk = 1$
(a)

DNA supercoiling described numerically by changes in the 'linking number' Lk.

The linking number is the most descriptive property of supercoiled DNA.



$Lk = 6$
(b)

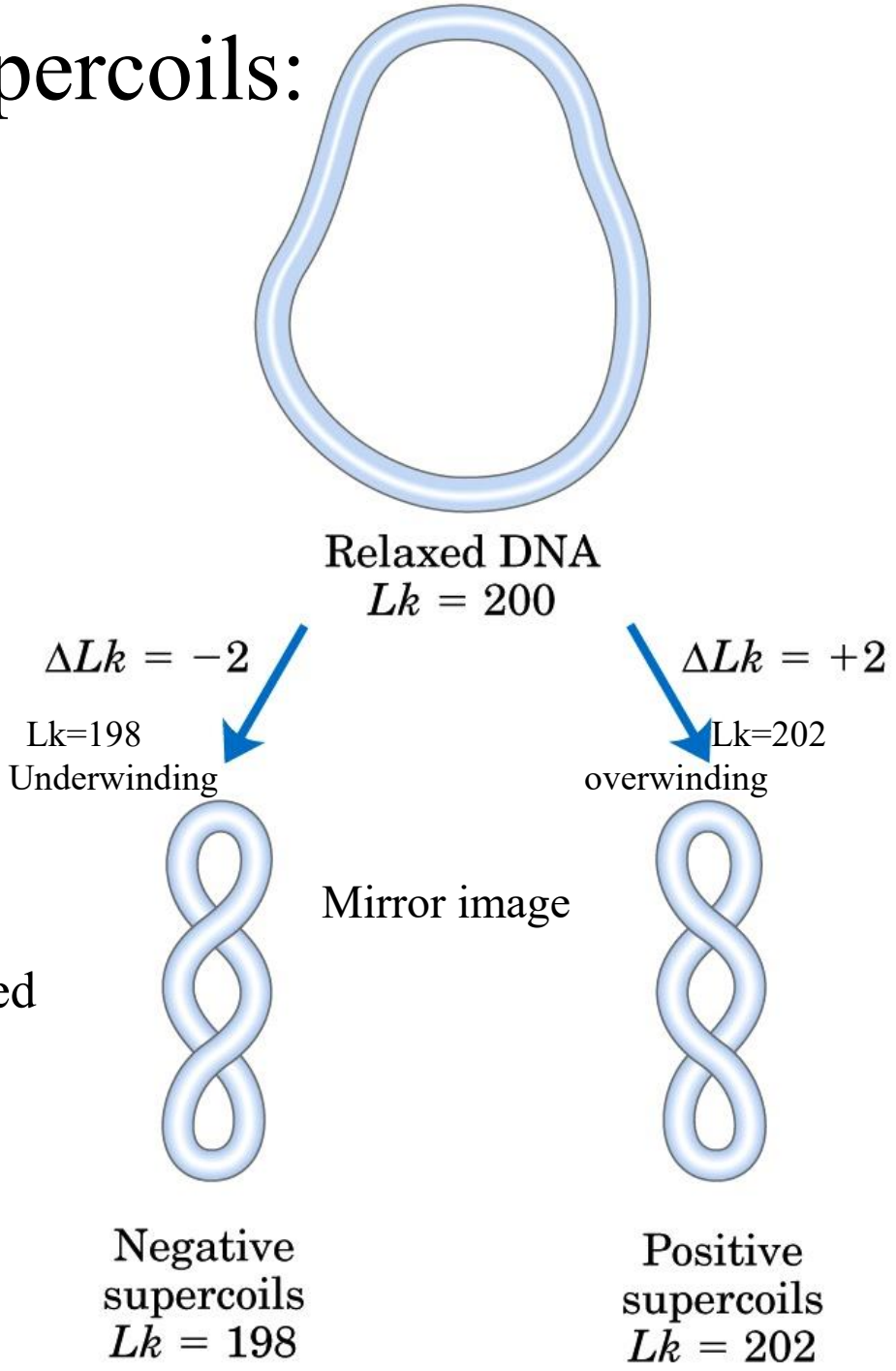
Negative and positive supercoils:

Negative supercoiling is naturally prevalent.

Negative supercoiling =
underwinding=left handed helix

Positive supercoiling =
overwinding= right handed helix

Negative supercoiling advantageous in replication→ its easier to open/unwind.
While positive supercoiling is more condensed



Topoisomerases

DNAs that differ only in the linking number (Lk) = **topoisomers.**

Enzymes that underwind/relax DNA = topoisomerases=

- The degree of supercoiling in the cell controlled by topoisomerases.

- Advantage: permits DNA to be transiently and locally melted to permit the enzymes of DNA replication and transcription to copy and synthesize new DNA or RNA.

- There are two classes of topisomerase:

  Type 1 topoisomerases

  Type 2 topoisomerases (DNA gyrase in Ecoli)

Topoisomerase Inhibitors:

Topoisomerases are essential enzymes:

topoisomerases → targets for antibiotics and other drugs.

novobiocin or nalidixic acid. Both are antibiotics that inhibit DNA gyrase. But not eukaryotic topoisomerases.

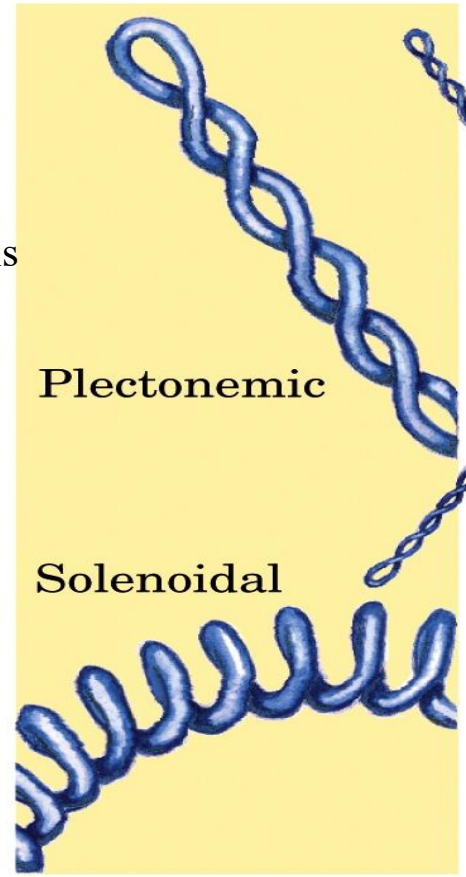
Eukaryotic topoisomerase inhibitors:

such as doxorubicin and etoposide, are used as chemotherapeutic agents in cancer therapy.

Forms of supercoiling

- **In prokaryotes**, plectonemic (twisted thread) supercoils are predominant, because of circular chromosome and small amount of genetic material.
- **In eukaryotes**, both present but solenoidal supercoiling most effective in compacting DNA.

Extended right handed coils



Plectonemic supercoiling does not provide sufficient compaction of DNA in the cell.

Much degree of compaction for the same DNA molecule

Tight left handed turns

(a)



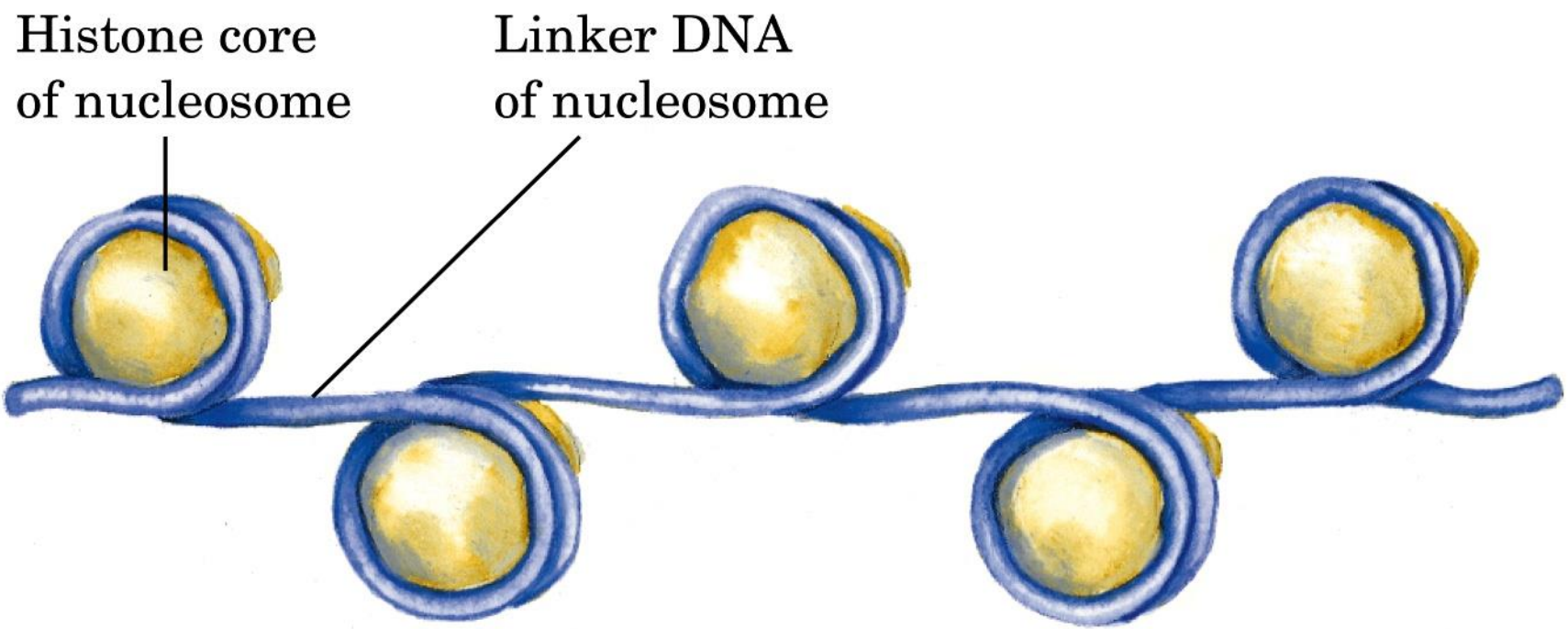
(b)

Beads on a string:

A core nucleosome = 2 copies of 4 histones - H2A, H2B, H3, H4 associate to form an octamer around which **146 bp** of DNA is wound in a **left-handed coil**.

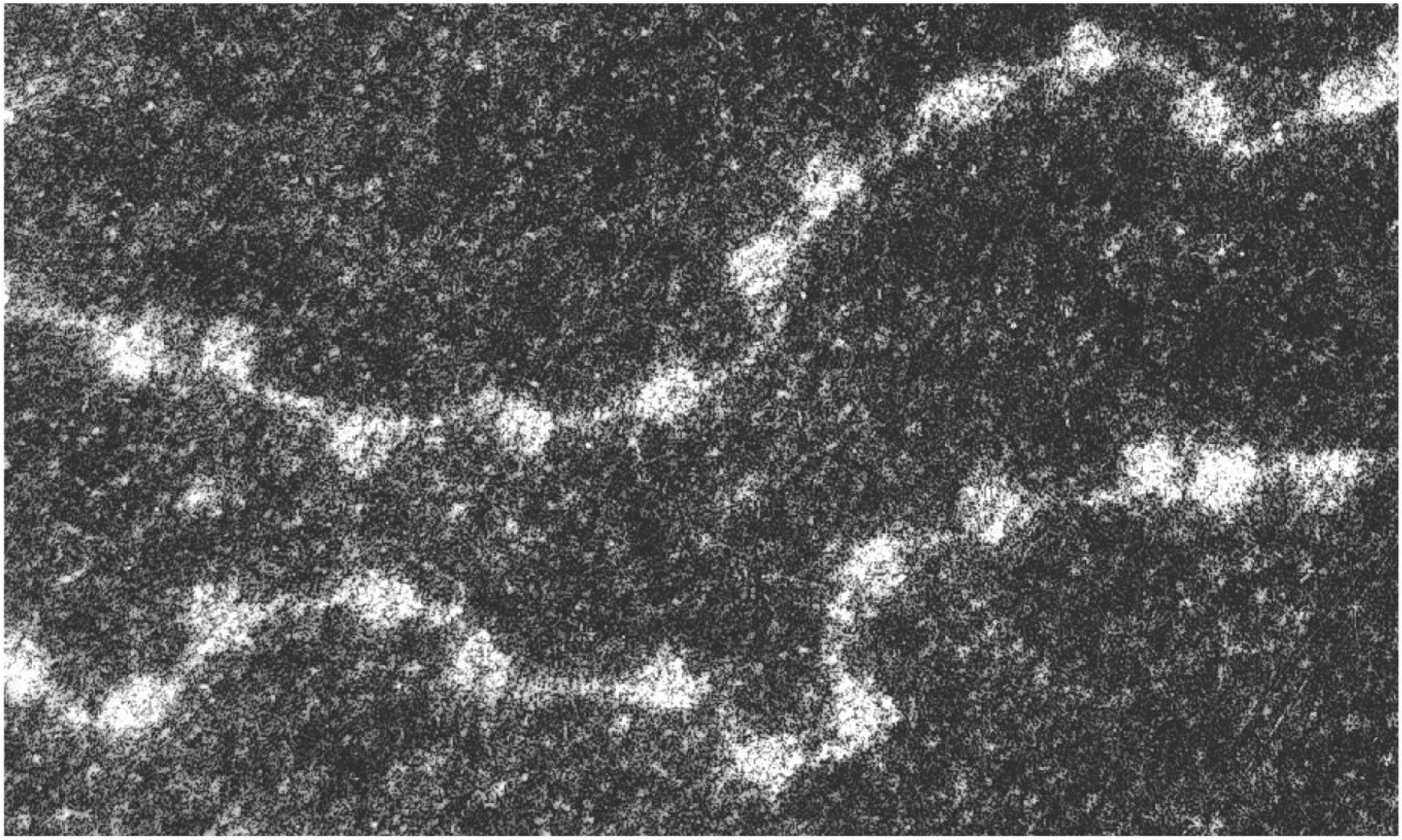
histone proteins rich in +ve charged amino acids (20-30% Arg + Lys) bind to DNA mostly via electrostatic interactions with the phosphodiester backbone of the DNA.

The remainder 200-146= 54 bp = linker DNA



(a)

Electron micrograph :
Nucleosomes, histone complexes bound to DNA



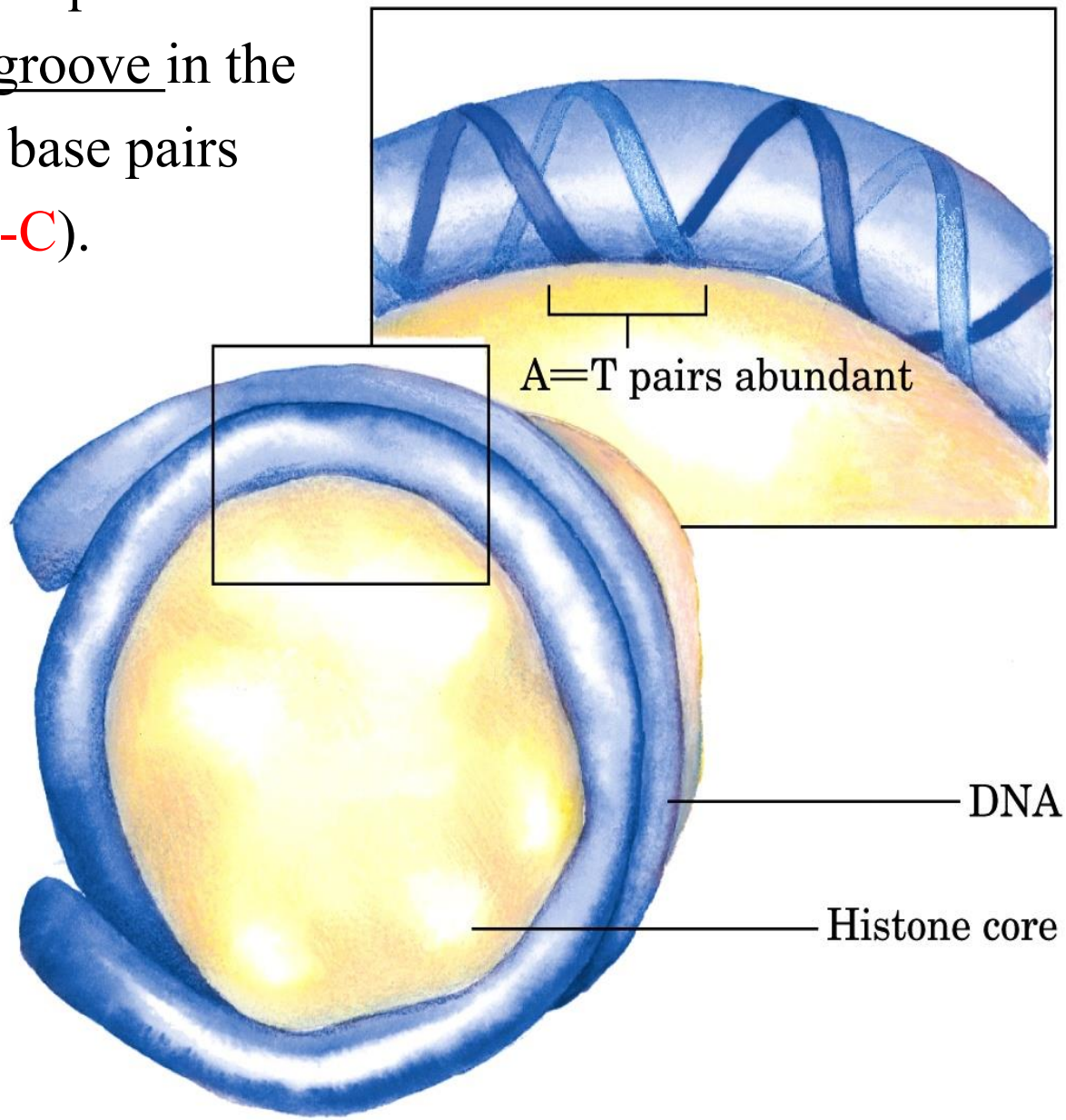
(b)

Nucleosomes two major purposes within the cell nucleus:

1- provide the level of compaction required to fit dsDNA into the cell nucleus.

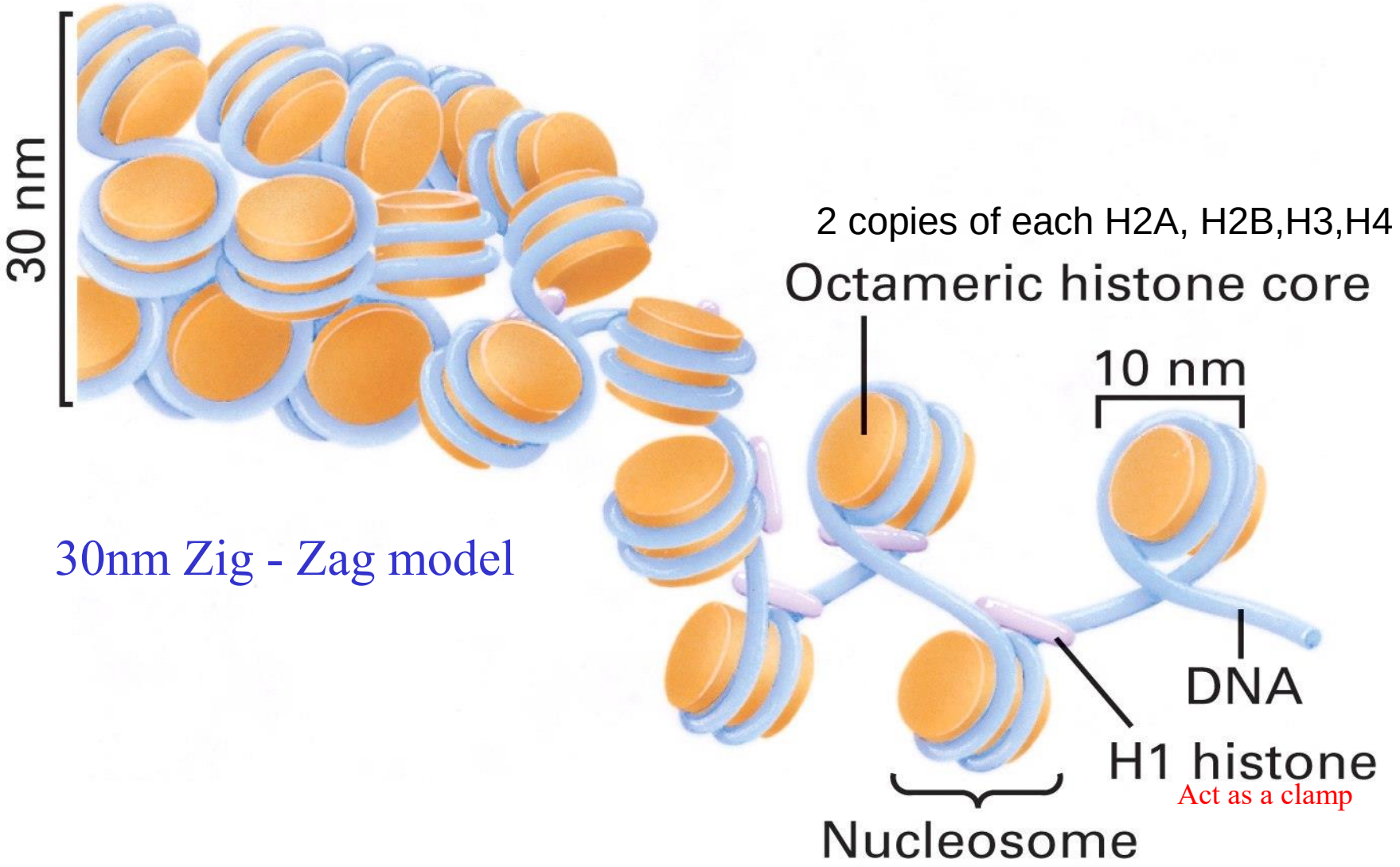
2- important in the regulation of transcription by preventing RNA pol from unnecessarily accessing the promoter regions of genes which are not needed by the cell. If the requirements of the cell change, enzymes known as remodeling factors can remove/change the position of the nucleosome to allow access.

Tight wrapping of DNA around the nucleosome's histone core requires compression of the minor groove in the helix at points rich in A=T base pairs (easier to compress than G-C).



Model of the 30nm condensed chromatin :

Each nucleosome associates with one H1 and the fiber coils into a solenoid structure with a diameter of 30nm.



30nm Zig - Zag model

- Five histones H1, H2A, H2B, H3, H4 rich +ve charged basic a.a. interact with -ve charged phosphate groups in DNA.
- Some modified post-translationally methylation (lysine), acetylation (lysine), phosphorylation (serine), →neutralizing / converting to –ve charge.

Histone Acetylation and Deacetylation

- In histone acetylation and deacetylation, the histones are acetylated and deacetylated on lysine residues in the N-terminal tail as part of gene regulation.
- "histone acetyltransferase" **(HAT)** or "histone deacetylase" **(HDAC**). The source of the acetyl group in histone acetylation = Acetyl-CoA, The acceptor of acetyl group in histone deacetylation is CoA.
- Acetylation brings a –ve charge, neutralise the +ve charge on the histones and decreases the interaction of N termini of histones with –ve charged phosphate groups of DNA. **the condensed chromatin is transformed into a more relaxed structure** which is associated with greater levels of gene transcription. This relaxation reversed by HDAC activity.
- Relaxed, transcriptionally active DNA =**euchromatin**.
- condensed (tightly packed) DNA =**heterochromatin**..

Several Orders of chromatin packing give to the compact chromosome.

